What is the correlation between age groups and their preferred game genres on PlayStation, and how does this affect the average time spent playing per week?

GROUP 5, Rochester Institute of Technology, USA

This study investigates the correlation relationship between PlayStation users, focusing on age groups, their preferred game genres on PlayStation, and the average time spent playing per week and developing the structured survey methodology on Google Forms as we gathered data from PlayStation users to collect data focusing on their game genre preferences and gaming habits. The study showed significant differences in gaming frequency and genre preferences across different age groups. Younger players (aged 18-25) demonstrated the most gaming frequency, averaging 15-20 hours per week, primarily engaging in action/adventure and multiplayer games. The research uncovered that living situations and game genre preferences impact gaming habits more than age and occupation. This study provided valuable insights into diverse gaming behaviors and preferences. The PlayStation users highlight the significant influence of age and gaming on frequency and genre choice.

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1 INTRODUCTION

This research paper explores the relationship between age groups, their preferred game genres on PlayStation, and their corresponding impact on their weekly playtime. We focused on observing how different age groups exhibit preferences and behaviors in the evolving gaming landscape. Inspired by Birk, Friehs, and Mandryk's findings, our study focuses on how gaming preferences align with age, from the performance-oriented to more choice and enjoyment center [1]. Additionally, influenced by Lee, Clarke, and Kim's research, we explored the variation in information-seeking behaviors across age and gender in the gaming context [2]. Those previous studies highlighted gaps in understanding how different demographic groups engage with games. Our research aims to fill those gaps by investigating how game genre preferences influence gaming frequency across various age groups. Our approach was a structured survey method to collect and analyze data, providing insights into the influence of game genre preferences on weekly gaming habits across different age groups. We designed this methodology to address our research questions directly, offering a perspective on the diverse gaming behaviors and preferences among PlayStation users.

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2 METHODS

2.1 Recruitment

We recruited the participants by sharing the Google Form survey link by sharing with our peers and posting it on our social media to reach out to our peers. We asked our peers if they played PlayStation regularly, and then we sent them a link to take Google Forms. We also posted on the Facebook group along with the PlayStation Trophy Hunters (75k members) and Deaf Gamer's Network (11.4k members). We also posted on Quora posts as a global post as well in a gaming-specific group with over 15K members and Reddit on a PlayStation server forum with over 5k members.

2.2 Survey Questionnaire

For our data collection, we utilized Google Forms and structured it into five sections. The initial section gathered preliminary qualifying information and six demographic questions about participants. The second section focused on the PlayStation-related queries with five questions about PlayStation. Crucially, if the respondents answered question 11 - "Have you played PlayStation at least once every two months in the last year?" - At this point, their survey will conclude there, and the survey will thank the participants for their involvement. The third section of our survey focused on identifying participants' game genre preferences with the five questions. The fourth section focused on time spent playing with twelve questions designed to gather detailed insights. Finally, the last section offered respondents an opportunity to provide additional comments to allow them to share any thoughts they have on gaming habits and preferences. In the concluding section, we included a brief message expressing our gratitude to the participants for contributing to the study. This gesture of appreciation is essential to acknowledge their time and effort in providing valuable data for our research. For a comprehensive view of our research questions, please refer to the appendix, where you will find a detailed list of the questions used in our survey. The survey should take no more than 10 - 15 minutes to complete.

2.3 Participants Demographic

We surveyed a total of 30 participants. We must eliminate four participants who voted "No" for not playing PlayStation at least once a week every two months in the last year. Twenty-six total participants selected "Yes" on this question. The participants included 18 males, four females, and four individuals, each identifying uniquely as non-binary, trans-masc nonbinary, gender fluid, and genderfluid trans. Among the participants, those aged 26-30 comprised 46.2%, or 12 participants, while those aged 18-25 comprised 30.8% or 8 participants. Additionally, there was 1 participant from each age group, 10-17 and 41-45, and 2 participants from each age group, 31-35 and 36-40.(M=27.17, SD=6.22) Regarding the living situation, 20% of the participants reported living alone, while 80% lived with others, including parents, spouses, or children.

2.4 Note on Participant Engagement and Response Data

We did not have the opportunity to track the total number of people who clicked on the link to our Google Form. As a result, we cannot provide specific data regarding the response rates, which will detail how many individuals we reached out to versus how many participated in the survey. We based our analysis and conclusion on the responses we received without the context of the overall engagement levels.

3 RESULTS

3.1 PlayStation Gaming History

Over 60% of the surveyed participants have played PlayStation for eight or more years, indicating a long-term engagement with the gaming platform, particularly from a young age. Moreover, a significant proportion of

the participants noted owning multiple gaming consoles (Figure 2). In addition, around 30% of the respondents reported not using PlayStation accessories. Interestingly, we observed the high popularity of Nintendo consoles among the participants (approximately 75%), which prompts further investigation to determine if there is a correlation between individuals with spouses or kids and ownership of Nintendo consoles.

3.2 Preferred Gaming Genres and Likelihood to Try New Games

The survey highlighted action, adventure, and RPG (role-playing games) as the preferred gaming genres among the participants, albeit with varying rankings. The sports genre showed an interesting pattern, with some participants considering it their top genre, while others did not prefer it. Additionally, a significant majority of participants, over 70%, expressed a likelihood to try new games within the same genre. This decision was influenced equally by similarity in the game played, price, and online reviews. Likewise, almost 40% of participants were ready to try new games in different genres, with 30% of them being neutral. Surprisingly, 65% of people trying a new genre would do so based on online reviews and recommendations by friends. These findings raise the prospect of delving deeper into the influence of social living situations on gaming habits, particularly regarding receptiveness to recommendations and reliance on online reviews for game selection.

3.3 Changes in Gaming Preferences Over Time

An assessment of changes in gaming preferences over time showcased a shift from playing more Nintendo and RPGs at a younger age to a greater interest in sports and FPS (first-person shooter) games as the participants grew older. This trend aligns with the evolving technology in video games, leading to increased genres and varieties available to gamers.

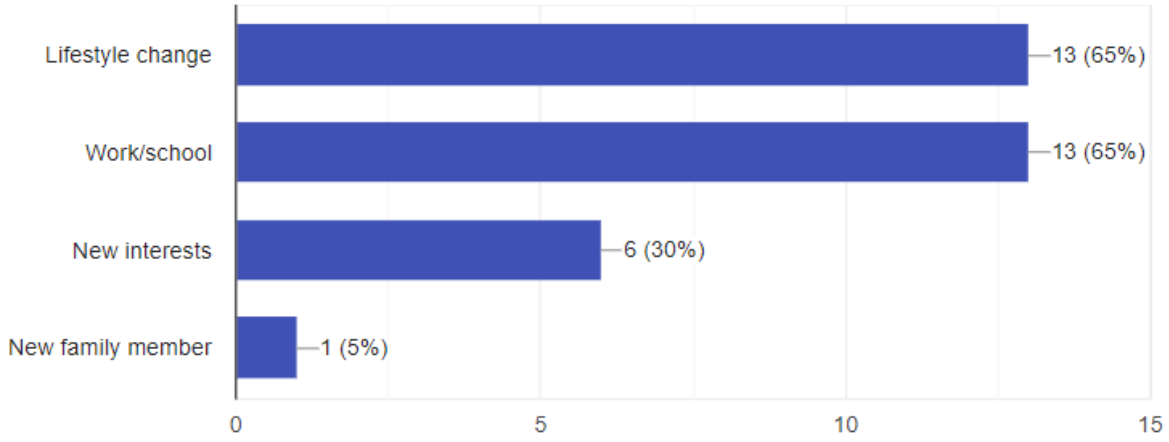


Fig. 1. Reasons for Changes in Gaming Habit

3.4 Gaming Habits and Patterns

An overwhelming majority, over 80%, of the participants reported engaging in gaming activities at least a few times a week, prompting a comparative analysis of this frequency across different survey distribution channels. The PlayStation 5 emerged as the preferred choice for gaming among approximately 75% of the participants.

Moreover, the study highlighted consistent gaming time patterns for most genres, paving the way for further investigations into playtime variations based on specific gaming genre preferences. The gaming patterns were observed equal for gaming in chunks or binge gaming. Notably, a significant proportion of participants indicated playing between 6 pm and early morning, with changes in playtime observed between weekdays and weekends. The factors affecting the most gaming hours were skill improvement and story progression for over 35

Furthermore, the study identified that a considerable portion predominantly engages in single-player gaming with almost 23% playing only multiplayer games and 40% engaging in very little multiplayer games. The players engaging in multiplayer games had a higher likelihood of competitive multiplayer gaming over cooperative play Over 65% of participants reported a change in their gaming habits in the past year. These changes correlate with the participants' occupations and lifestyle factors. Figure 1 shows a clear response of the participants.

4 DISCUSSION

The original research question asked about relationships between gamer demographics, preferred gaming genres, and a gamer's playtime. We saw multiple variables from the results and chose to analyze the data in a few different ways, including a) the impact of lifestyle and b) the impact of genre preferences. In this survey, a participant's lifestyle includes their age group, living situation (i.e., alone or with others), and occupation. The results and insights must also be viewed from the research method's perspective. We sent the survey to various online gaming communities and forums, and survey participation was at the viewer's discretion. As such, there is likely a bias for participants to skew towards being active and passionate gamers as opposed to casual gamers and for participants to be younger and in online communities.

4.1 Impact of Player Demographic

After analyzing the data, the gaming frequency varied significantly across different age groups. The younger demographic, specifically those aged 18-25, exhibited the highest gaming frequency, spending an average of 15-20 hours per week engaging in PlayStation games. Conversely, the older age brackets displayed a decrease in gaming frequency, with individuals aged 46 and above spending approximately 5-10 hours per week on gaming activities. When looking at the habits of those who live alone (20% of participants) compared to those who live with others (80% of participants), they are not more prone to playing only single-player games or games of any specific genre. However, their playtime habits differ: participants who live alone are more likely to play more hours, and they all play games every day. 100% of participants who lived alone played at least 10 hours/week, with 50% playing more than 20 hours. In contrast, only 20% of participants who lived with others played more than 10 hours/week, and 10% played 20+ hours/week. This is not surprising, as people who live alone have more free time at home, whereas those who live with others may have to split game time with other social activities or care for their family. A participant's occupation did not seem to influence gaming habits strongly.

The survey also delved into the factors influencing gaming habits, encompassing aspects such as work-life balance, technological advancements, and social interaction. The analysis revealed that younger participants emphasized gaming as a means of socializing and unwinding, while the older age groups perceived gaming as a leisure activity intertwined with technological advancements.

4.2 Impact of Player Genre Preferences

In addition to the participants' lifestyle, an analysis of their genre preferences also impacted their gaming habits. Certain age groups leaned towards specific genres. The younger demographic demonstrated a strong inclination towards action-adventure and multiplayer games, whereas the older age brackets displayed a predilection for simulation and strategy games. This disparity in gaming preferences highlights the influence of age on the choice of gaming genres. In particular, 87% of participants whose top game genre is Sports played < 5 hours/week,

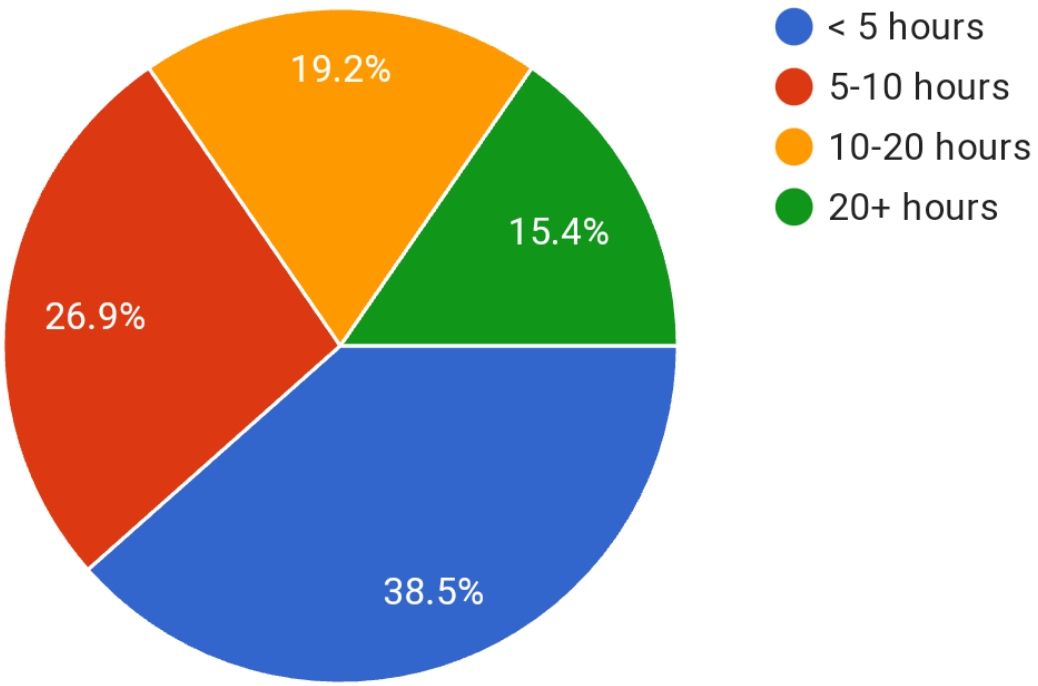


Fig. 2. Gaming time duration over the week

compared to the average response where 38% of participants played < 5 hours/week. Fans of Sports games were also more likely to spend time gaming to improve their skills: 62% of these participants had skills improvement as their primary driver, compared to other participants, where only 16% responded with skill improvement as their main driver.

5 CONCLUSION

This paper attempts to understand the relationship between gamers and their gaming habits. Based on the results, there seems to be some correlation between a participant's age and living situation and their play time; namely, younger players and players who live by themselves are more likely to play more hours per week. This seems true regardless of the player's occupation and preferred game genres. Additionally, younger players are more inclined towards action and multiplayer games, while older players lean towards strategy and simulation genres. Further, a player's preferred genre of games influences both their playtime and motivations for play. Participants who like sports games play fewer hours per week than those who prefer other genres, such as adventure and RPG and are also motivated primarily by skill improvement rather than story progression and level grinding.

6 FUTURE WORK

The survey provided valuable insights into PlayStation users' gaming habits. We gained insights and ideas from this research on a small sample of volunteer participants from various online forums. Future work should include participants who do not frequent these forums and target those of a larger age group. During data analysis, there was participant feedback on the open-answer questions that the research team had yet to consider in the survey, such as the impact of having a PlayStation Plus Membership and the utility of other PlayStation accessories, such as a JoyStick or VR headset. A follow-up survey could include more questions about these topics. Finally, this survey relies on a participant's ability to self-report accurately. We could gain further insights by comparing self-reporting data to empirical play time data available online directly from PlayStation (Sony), Nintendo, and other platforms. To enhance the comprehensiveness of the findings, we also suggest that the results be compared with existing research papers to identify similarities and differences in the obtained outcomes.

REFERENCES

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7 APPENDIX

7.1 Survey questionnaire

Survey questionnaire

Intro:

Welcome to the PlayStation Gaming Preferences Survey!

Are you a PlayStation enthusiast? We'd love to hear from you! We're conducting a study to explore the fascinating world of gaming habits. We focus on understanding how different age groups prefer various game genres on PlayStation and how this influences weekly gaming time.

Your insights are invaluable! This survey consists of questions about your gaming preferences, habits, and the types of games you enjoy most on your PlayStation console. It should take approximately 10-15 minutes to complete. Rest assured, your responses will be kept confidential and used only for research purposes.

Pre-Qualifying Questions

DEMOGRAPHIC PAGE - Section 1

(1) What is your first name or initials? (Optional)
    a. Text field
(2) What is your gender identity? Radio Choice
    a. Male
    b. Female
    c. Transgender Man
    d. Transgender Woman
    e. Non-binary
    f. Other (Short Answer)
(3) What is your current age? Radio Choice
    a. 10-17
    b. 18-25

c. 26-30
d. 31-35
e. 36-40
f. 41-45
g. Others (short answer)

(4) What is your occupation? Multi-choice
a. Full-Time Job
b. Part-Time Job
c. Student
d. Unemployed
e. Retired
f. Self-Employ/Freelancer
g. Others (short answer)

(5) Who do you live with? Multi-Choice
a. Alone
b. Live with parent
c. With spouse/partner
d. With children
e. With roommates
f. Others (short answer)

(6) How often do you play games alone or with others?
a. Alone
b. With Friends online
c. With Family
d. With Friends in Person
e. Others (short answer)

—— Page change —– Section 2
Console Demographic

(7) Do you own a Playstation console? (Yes/no)

(8) How long have you owned/played PlayStation? Radio Choice
a. < 3years
b. 3 - 8 years
c. 8+ years

(9) Select all the PlayStation consoles that you are currently playing (multiple selections allowed):
a. PlayStation 5
b. PlayStation 4
c. PlayStation 3
d. PlayStation Vita
e. PSP
f. Others (short answer):

(10) Do you own/use any of the PlayStation accessories?
a. Joystick
b. Steering Wheel
c. VR headset

d. Move (Motion sensor)
e. Other (short answer)

(11) Have you played PlayStation at least once every two months in the last year?
a. Yes
b. No *

(12) Do you own/play other gaming consoles/platforms?
a. Xbox
b. Nintendo Switch
c. PC/MAC
d. I do not play with other gaming consoles/platforms

*If the respondents choose question 11, option 'no' ends the survey and expresses gratitude for their participation.

Section 4 - Not playing PS: A big thank you for your time and effort in completing our survey. We are specifically looking for insights from active players, and your active involvement in the PlayStation community is what makes this research meaningful and impactful.

—— Page change —– Section 3 - Game Genre Preference

Main Survey Questions

Section 3: Game Genre Preferences

(1) What are the top 3 genres you most frequently play on PlayStation?(multiple selections allow as Top 1, Top 2, Top 3)
a. Action
b. Adventure
c. RPG (Role-Playing Game)
d. Sports
e. Simulations
f. Strategy
g. Puzzle
h. Horror
i. Other - (Short answer)

(2) How likely are you to try a new game in the same genres you already play?
a. Scale
i. Likely - not likely
b. What factors into this decision?
i. Similarities to games already played
ii. Price
iii. Online Reviews
iv. Friend recommendations
v. Other (short answer)

(3) How likely are you to try a new game genre?
a. Scale
i. Likely - not likely
b. What factors into this decision?
i. Similarities to games already played
ii. Price
iii. Online Reviews

iv. Friend recommendations
v. Other (short answer)

(4) Have your game genre preferences changed over the years?
a. Yes
b. No

(5) If yes, how have they changed?
a. Open-ended

—— Page change —– Section 4 - time spend playing
Section 4: Time Spend Playing

(6) How often do you play your PlayStation? Radio Choice
(a) Rarely (a few times a year)
(b) Occasionally (a few times a month)
(c) Frequently (a few times a week)
(d) Everyday

(7) Which console have you played mainly for the past two to three years? Radio Choice
(a) PlayStation 5
(b) PlayStation 4
(c) PlayStation 3
(d) Others (Short Answer)

(8) How many hours do you spend playing PlayStation per week?
a. < 5 hours
b. 5 - 10 hours
c. 10 - 20 hours
d. 20+ hours

(9) Is more of your time spent playing games:
a. A few long stretches of time (ie multi-hour binges on the weekend)
b. Lots of small chunks of time (ie an hour after work each day)

(10) Does the amount of time you spend playing vary by genre?
a. Yes/No
b. If Yes, elaborate (short answer)
i. Add multiple choice for the genre(s) with the most hours spent.

(11) What time of day do you most frequently play PlayStation games?
a. Morning (6am - 12pm)
b. Afternoon (12pm - 6pm)
c. Evening(6pm - 10pm)
d. Late night(after 10pm)

(12) Does your gaming time change during weekends compared to weekdays?
a. Yes
b. No

(13) What factors of a game result in the most hours spent playing?
a. Level grinding
b. Skills improvement
c. Story progression

d. Social/Competition with others
e. Other (short answer)

(14) What is the ratio of time spent on single-player vs multiplayer games?
a. 100% single player
b. 80-20
c. 60-40
d. 50-50
e. 40-60
f. 20-80
g. 100% multiplayer

(15) If you play multiplayer games, what is the ratio of gaming time for cooperative games vs competitive games?
a. 100% cooperative
b. 80-20
c. 60-40
d. 50-50
e. 40-60
f. 20-80
g. 100% competitive
h. I do not play multiplayer games

(16) Have the time you spend playing PlayStation games changed in the last year?
a. Yes/No

(17) If yes, what are the main reasons for this change?
a. Lifestyle change
b. work/school
c. New interests
d. New family member
e. Others (short answer)

Section 5: Additional Comments

18. Please share any additional thoughts you have about your gaming habits and preferences.

a. Short answer

End of section:

Thank You for Your Participation! We greatly appreciate you taking the time to complete this survey. Your input is invaluable and plays a crucial role in our research. Thanks to your contribution, we're one step closer to understanding the diverse preferences and habits of the PlayStation gaming community.

ChatGPT generated: 02/27/2024 - 3:34 PST using ChatGPT.

Input: Can you make it like media content to recruit on the Facebook group related to the PlayStation survey? Why? We want to state that we can recruit our participants to take this survey.

Output:

Hey, PlayStation Lovers!

We're on a quest to uncover the gaming habits of PlayStation players across various age groups and how these preferences impact your weekly gaming time. And guess what? You can be a part of this exciting journey!

Be a Part of Our PlayStation Gaming Preferences Study!

Whether you enjoy action-packed adventures, strategic puzzles, or immersive RPGs, your gaming experiences

are crucial to our study. This is more than just a survey; it's a platform for you to share your gaming passion and how it integrates into your life.
https://forms.gle/XbyKv6G8mRgFA5Jg8 Google Form
Why Does Your Participation Matter?
Contribute to a community-driven study on PlayStation gaming trends.
Express your gaming journey and preferences.
Elevate your voice within the gaming community.
Help us gain valuable insights into the diverse world of gamers.
It's easy, engaging, and takes just 10-15 minutes. Rest assured, your responses will remain confidential and are integral in shaping our understanding of the PlayStation gaming landscape.
Thank you for participating in our survey!
works for me

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